



Power Electronics (EE3032)

Final Exam

Date: May 22, 2025

Course Instructor(s)

Dr. Muhammad Naeem

Total Time (Hrs): 3

Total Marks: 80

Total Questions: 5



Roll No

Section

Student Signature

Do not write below this line

Attempt all the questions.

CLO 3: Design power switches.

Q1: Design a p-channel high-side switch for load resistance $R_{load} = 10 \Omega$. Determine the voltage at each node of the designed switch when $e_{in} = 2.4$ V. Determine the load voltage v_{load} if IRF9530 on resistance is 0.57Ω . If the switching frequency is 30 kHz and the duty cycle is 80%, calculate (a) Power delivered to the load, (b) power dissipated in IRF9530, and (c) the required heat-sink maximum thermal resistance for $T_A = 50^\circ\text{C}$. ($\theta_{JC} = 1.7^\circ\text{C/W}$, $\theta_{CS} = 2^\circ\text{C/W}$) [10]

CLO 4: Design power converter circuits.

Q2: Design a three-phase uncontrolled bridge rectifier that delivers $I_{dc} = 30$ A to a three-phase resistive load $R_{load} = 10 \Omega$. Determine the volt-ampere rating of the transformer secondary windings. Sketch the input and output voltage waveforms. Also calculate the rectification efficiency, form factor, ripple factor, transformer utilization factor, power factor, peak current through each diode, and peak inverse voltage of each diode. [20]

CLO 4: Design power converter circuits.

Q3: Design a buck regulator using LM2595 IC to convert an unregulated voltage of 18 V dc to 5 V dc. Use the output resistor of 10 Ω . Other design parameters are: $f_{osc} = 100$ kHz, $\Delta v_{pp} = 0.1$ V and $\Delta i_{pp} = 20\%$ of the load current, $I_{quiesc} = 10$ mA and $V_{sat\ max} = 1$ V for LM2595. Return the following parameters as part of the designed circuit: (Note: Circuit diagram not required in this question) [20]

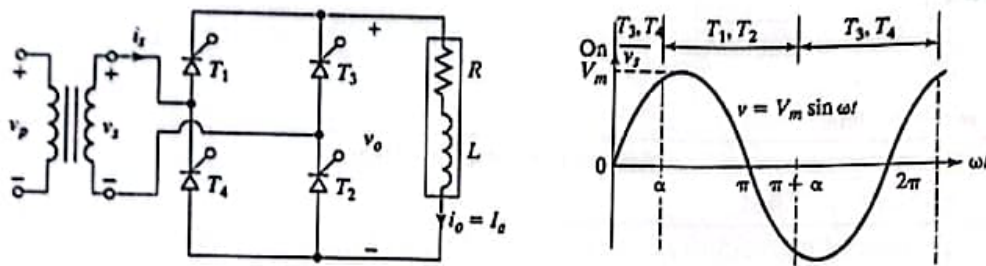
- Inductance of the inductor required
- Peak current rating of inductor
- Capacitance of the input capacitor
- Current rating of input capacitor
- Voltage rating of the input capacitor
- Capacitance of the output capacitor
- Resistances of feedback resistors R_l and R_f
- Equivalent Series Resistance of the regulator
- Power dissipated by the LM2595 IC
- The output power delivered to the load
- The input power of the regulator



CLO 5: Analyze thyristor-based circuits.

Q4: Analyze the single-phase controlled rectifier shown below. The transformer secondary voltage v_s is also plotted below in which $\alpha = \pi/3$ is the firing angle. The fundamental component of the input current is $0.9I_a$. Sketch the output voltage v_o if the load is highly inductive. Also calculate: [10]

- i. The average dc output voltage V_{dc}
- ii. The rms output voltage V_{rms}
- iii. The harmonic factor of the input current
- iv. The displacement factor
- v. Input power factor



CLO 5: Analyze thyristor-based circuits.

Q5: Analyze the thyristor-based circuit shown below. It is operated by a three-phase Y-connected, 208 V, 60 Hz supply and the load resistance is 20Ω . It is required to obtain an average output voltage of 86.6% of the maximum possible output voltage, calculate [20]

- i. The firing angle α
- ii. The rms output voltage V_{rms}
- iii. The rms output current I_{rms}
- iv. The average output voltage V_{dc}
- v. The average output current I_{dc}
- vi. The average current through T_1
- vii. The rms current through T_1
- viii. The rectification efficiency
- ix. Transformer Utilization Factor
- x. Input power factor

